

14 — Bulk Carrier: L3 Variation from Vessel-Type Lookup

Proves · The Level 3 variation looked up from the vessel type when the field is left empty.

Mechanics this scenario turns on

- When the Level 3 variation field is empty the backend looks it up from the vessel type (Bulk 250 · Container 1 500 · LNG 1 000 · otherwise the 500 default), all from `appsettings.json`.
- Level 3 (DRC) damps the hotel/mission swing: $\text{variation} \times 0.8$, minus whatever the battery already shaved, so the same kilowatt is never counted twice.
- Baseline rule: **no battery** → **the worst combination** (`count - 1`); **battery active** → **the third from worst** (`Math.Max(0, count - 3)`). It models what the ship is assumed to do today.

Panels described below · The point of the test: the L3 lookup · Power Demands · Baseline & IL1 — an honest zero

Anything not described here behaves exactly as in `01-excel-baseline` — same plant, same hours; only what this scenario changes is worked through below.

Trust · verified against the reference workbook. These figures are proof.

Read after · scenario 11.

Small bulk carrier from the DB row: curve 12 kn → **1 365 kW**, SM **20 %**, ME 1×12 000 (Diesel 4-stroke Medium), SG 200, AE 2×500. Profile modes: Transit 5 717 h/165 · Port 2 592/110 · Anchor 451/180 · Maneuvering 175 (400/190). No battery. **Load Variation field left EMPTY.**

The point of the test: the L3 lookup

```
HotelLoadVariationKw = empty → backend looks up VesselTypeName:
"Bulk Carrier 10,000 dwt" contains "Bulk Carrier" → appsettings VesselVariations → ±250
DRC reduction 20 %: 250 × 0.8 = 200
IL3 Optimization Details must read: "Variation: ±250 kW → ±200 kW" ← the proof
```

(Config map: Bulk Carrier 250 · Container 1500 · LNG 1000 · default 500.)

Power Demands

Transit propulsion = $1\,365 \times 1.20 = \mathbf{1\,638}$. All hotels via SG 200 (AE = 0 everywhere — the SG-forced rule again; the 12 MW ME idles at ~1 % in port/anchor to spin a 110–200 kW hotel). Energies: Transit $(1\,638 + 165) \times 5\,717 = \mathbf{10\,307\,751}$ · Port $110 \times 2\,592 = \mathbf{285\,120}$ · Anchor **81 180** · Maneuvering $(400 + 190) \times 175 = \mathbf{103\,250}$. Total hours 8 935.

Baseline & IL1 — an honest zero

Transit has exactly **one** valid combination (SG 200 covers hotel 165; both 500-kW AEs would sit idle → rejected). One row ⇒ baseline = optimal ⇒ **IL1 = Baseline = 1 883.4 t/yr, savings 0 t / \$0** against a \$110k investment.

The calculator refuses to invent value where a single-engine plant offers no choices. L3 also nets 0 at these load points (SG capacity 200 clamps the spike cycle); the lookup display is the deliverable, not the savings.

Takeaway: VesselTypeName's only direct role in the math is this L3 lookup — and the UI shows it verbatim in the IL3 details line.

KSailCalc manual test scenarios — calculation walkthrough · regenerated 2026-08-10 · numbers verified against commit 559aef2 and unchanged by the backend/client refactors (18 golden snapshots byte-identical)